FOUNDATION LOAD : PIPE RACK–1

SL No	Description	BP1			BP2			BP3			BP4			BP5			BP6			BP7			Remarks
		X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	
1	DEAD LOAD	0.0	-22.4	0.0	0.0	-24.2	-0.2	0.0	-48.3	-0.2	0.2	-39.6	-0.2	0.0	-35.5	0.2	0.0	-20.2	0.0	0.0	-7.1	0.2	A. ALL LOADS FURNISHED IN THIS TABLE ARE UNFACTORED ACTING LOADS. B. PIPE LOAD SHALL BE CONSIDERED UNDER DEAD LOAD CATEGORY. C. FOR REVERSE DIRECTION OF LOAD CASE 6 TO 11, MULTIPLY ALL VALUES WITH -1.
2	PIPE EMPTY LOAD	0.2	-40.2	0.0	0.2	-9.1	0.0	-0.2	-38.8	0.0	-0.2	-24.5	0.0	0.0	-28.9	0.0	0.0	-17.9	0.0	-0.4	-12.3	-0.2	
3	PIPE OPERATING LOAD	0.2	-46.9	0.0	0.2	-15.3	0.0	-0.2	-47.2	0.0	-0.2	-31.9	0.0	0.0	-41.1	0.0	0.0	-27.9	0.0	-0.6	-18.6	-0.2	
4	PIPE TEST LOAD	0.2	-46.9	0.0	0.2	-15.3	0.0	-0.2	-47.2	0.0	-0.2	-31.9	0.0	0.0	-41.1	0.0	0.0	-27.9	0.0	-0.6	-18.6	-0.2	
5	LIVE LOAD	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.7	0.0	
6	PIPE LOAD +X	4.0	16.3	-0.2	4.0	-16.3	0.2	6.6	29.1	0.0	6.6	-28.9	0.0	8.1	45.0	0.0	8.1	-44.7	0.0	4.4	52.0	-11.2	
7	PIPE LOAD +Z	0.4	2.4	0.5	0.4	-2.2	0.5	0.2	0.5	0.5	0.2	0.0	0.5	-0.2	-1.1	0.7	-0.2	0.6	0.6	-0.2	-24.6	9.2	
8	WIND LOAD +X	25.0	142.2	0.5	25.0	-141.8	-0.5	29.4	178.0	0.5	29.4	-177.0	-0.4	26.9	159.9	0.3	26.9	-157.2	-0.3	11.4	38.0	1.4	
9	WIND LOAD +Z	0.7	1.7	1.9	0.7	-5.4	1.8	0.2	-4.1	2.0	0.2	-4.8	1.8	-0.4	5.1	2.2	-0.4	7.7	2.0	-0.3	-72.5	29.5	
10	SEISMIC LOAD +X	1.1	5.9	0.0	1.1	-5.9	0.0	1.1	7.0	0.0	1.1	-6.9	0.0	1.0	6.5	0.0	1.0	-6.4	0.0	0.4	1.6	0.2	
11	SEISMIC LOAD +Z	0.0	0.2	0.0	0.0	-0.3	0.0	0.0	-0.2	0.0	0.0	-0.3	0.0	0.0	0.3	0.0	0.0	0.4	0.0	0.0	-4.4	1.7	
12	TEMPERATURE LOAD +ve	0.0	3.6	2.1	1.3	-4.0	2.5	-1.4	-11.9	1.3	-0.2	6.3	1.5	-0.8	4.8	-0.2	0.5	1.3	0.3	0.3	1.9	1.7	
13	TEMPERATURE LOAD -ve	0.2	-1.3	-1.0	-0.6	0.7	-1.1	0.5	5.2	-0.6	-0.3	-0.6	-0.7	0.4	-1.6	-0.4	-0.4	-2.4	-0.4	-0.2	-1.3	-0.8	

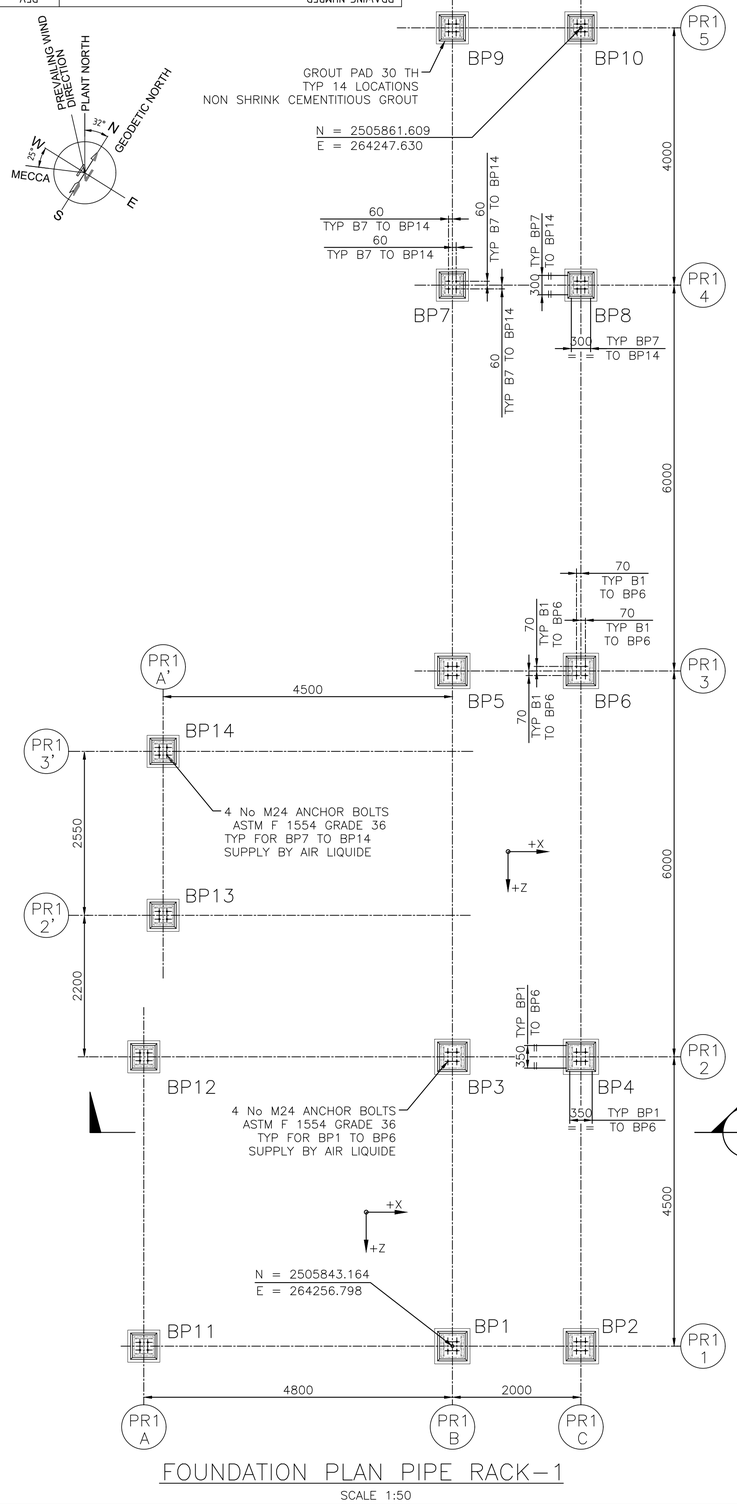
SL No	Description	BP8			BP9			BP10			BP11			BP12			BP13			BP14			Remarks
		X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	X (kN)	Y (kN)	Z (kN)	
1	DEAD LOAD	0.0	-8.7	0.9	0.0	-9.2	-0.7	-0.2	-10.9	0.0	0.0	-5.7	0.2	0.0	-5.7	-0.2	0.0	-20.8	0.0	0.0	-20.4	0.0	A. ALL LOADS FURNISHED IN THIS TABLE ARE UNFACTORED ACTING LOADS. B. PIPE LOAD SHALL BE CONSIDERED UNDER DEAD LOAD CATEGORY. C. FOR REVERSE DIRECTION OF LOAD CASE 6 TO 11, MULTIPLY ALL VALUES WITH -1.
2	PIPE EMPTY LOAD	0.4	-11.2	0.2	0.0	-22.7	-0.2	0.0	-12.0	0.2	0.0	-11.6	1.7	0.0	-11.6	-1.7	0.0	-1.0	0.0	0.0	-2.6	0.0	
3	PIPE OPERATING LOAD	0.5	-17.3	0.2	0.0	-30.4	-0.2	0.2	-18.5	0.2	0.0	-18.1	2.5	0.0	-16.8	-2.5	0.0	-1.3	0.0	0.0	-3.7	0.0	
4	PIPE TEST LOAD	0.5	-17.3	0.2	0.0	-30.4	-0.2	0.2	-18.5	0.2	0.0	-14.7	1.8	0.0	-12.5	-1.8	0.0	-1.3	0.0	0.0	-3.7	0.0	
4	LIVE LOAD	0.0	0.7	0.0	0.0	-6.2	0.0	0.0	-6.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
5	PIPE LOAD +X	4.4	-51.9	11.2	5.5	-0.6	-11.2	5.5	0.6	11.2	0.0	0.0	0.0	0.0	0.0	0.0	1.0	-0.4	0.0	0.8	-0.4	0.0	
5	PIPE LOAD +Z	-0.2	-20.3	8.1	-0.3	23.0	9.3	-0.3	21.8	8.1	0.0	-16.4	8.8	0.0	16.4	8.8	0.0	0.0	0.0	0.0	0.0	0.0	
6	WIND LOAD +X	11.4	-38.1	-1.4	11.3	44.2	1.6	11.3	-44.0	-1.5	1.5	0.0	0.0	1.5	0.0	0.0	3.7	-1.7	0.0	4.1	-2.4	0.0	
7	WIND LOAD +Z	-0.3	-66.8	28.2	-0.4	70.1	29.5	-0.4	69.1	28.2	0.0	-11.7	7.7	0.0	11.7	7.7	0.0	-76.6	18.2	0.0	76.6	18.2	
8	SEISMIC LOAD +X	0.4	-1.6	-0.2	0.4	1.9	0.2	0.4	-1.9	-0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.2	-0.2	0.0	0.2	-0.2	0.0	
9	SEISMIC LOAD +Z	0.0	-4.0	1.6	0.0	4.3	1.7	0.0	4.1	1.6	0.0	-0.6	0.4	0.0	0.6	0.4	0.0	-1.6	0.4	0.0	1.6	0.4	
10	TEMPERATURE LOAD +ve	0.6	7.5	-2.1	1.6	1.4	-1.6	-1.1	-10.7	-5.3	0.0	0.0	0.3	0.0	0.0	-0.3	-0.5	0.2	0.5	-0.3	0.0	-0.5	
11	TEMPERATURE LOAD -ve	-0.4	-4.0	1.1	-0.9	-0.7	0.9	0.6	5.9	2.8	0.0	0.0	-0.2	0.0	0.0	0.2	0.6	-0.2	-0.3	0.5	-0.2	0.3	

NOTE: PIPE LOAD +X & PIPE LOAD+Z ARE THE HORIZONTAL LOADS GENERATED DUE TO PIPE STRESS.
THESE TWO LOADS SHALL BE CONSIDERED ACTING SIMULTANEOUSLY IN OPERATING CONDITION.

NOTES

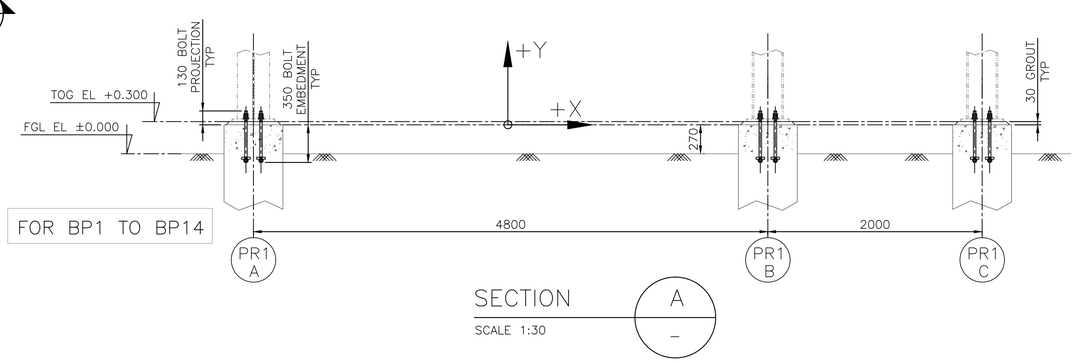
1. ALL DIMENSIONS ARE IN MILLIMETER.
2. ALL LEVELS ARE IN METER.
3. FGL LEVEL ±0.0 CORRESPONDS TO RL 912.350.
4. THE LOADS INDICATED IN LOAD DATA TABLE ARE ACTUAL. FOUNDATION DESIGNER SHOULD CONSIDER NECESSARY MARGIN.
5. SIZE OF FOUNDATION AND PEDESTAL SHALL BE DECIDED BY CIVIL DESIGN ENGINEER AS PER GIVEN LOADING.
6. ADDITIONAL REBAR REQUIREMENT AROUND ANCHOR BOLTS TO ENSURE NO FAILURE OF CONCRETE SHALL BE DECIDED BY CIVIL DESIGN ENGINEER AS PER GIVEN LOADING.
7. WIND & SEISMIC LOADS ARE AS PER SBC 301.

LEGEND :
CL – CENTER LINE
CG – CENTER OF GRAVITY
CONC – CONCRETE
EL – ELEVATION
EQP – EQUIPMENT
FGL – FINISHED GROUND LEVEL
Gr – GRADE
MPa – MEGA PASCAL
mm – MILLIMETER
No – NUMBER / NUMBERS
SQ – SQUARE
TH – THICKNESS
TOC – TOP OF CONCRETE
TOG – TOP OF GROUT
TOP – TOP OF SHIM PLATE
TYP – TYPICAL



FOUNDATION PLAN PIPE RACK–1

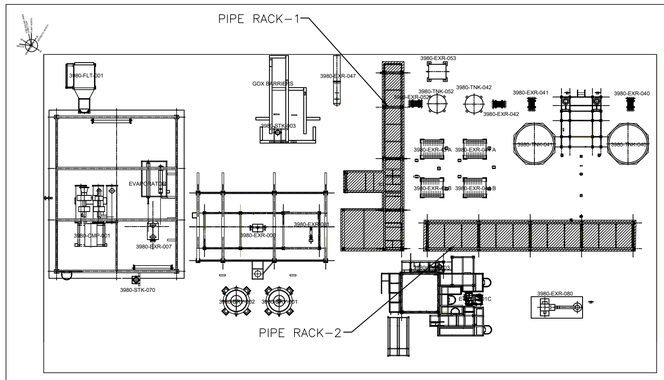
SCALE 1:50



FOR BP1 TO BP14

SECTION

SCALE 1:30



KEY PLAN

REV	DESCRIPTION	DRN	CHK	ENG	APP	EMP	DATE
D01	AS-BUILT	--	STY/SYC/SHH	--	BDA/PRI/JPD		22/02/2023
C01	ISSUED FOR CONSTRUCTION & REVISED AS PER EMPLOYER'S COMMENT ON 14/07/2020	--	STY/SYC/SHH	--	BDA/PRI/JPD		18/08/2020
A01	ISSUED FOR INFORMATION	--	STY/SYC/SHH	--	BDA/PRI/JPD		30/05/2020

DOCUMENT NUMBER | REFERENCE DOCUMENTS

Air Liquide ENGINEERING & CONSTRUCTION

AIR LIQUIDE CONFIDENTIAL INFORMATION: THIS PRINT IS THE PROPERTY OF L'AIR LIQUIDE S.A. OR ITS DIRECT AND INDIRECT) AFFILIATES OR SUBSIDIARIES AND CONTAINS VALUABLE CONFIDENTIAL INFORMATION. COPYRIGHT AND ALL OTHER RIGHTS IN THIS PRINT ARE EXPRESSLY RESERVED. NO PART OF THIS PRINT MAY BE DISCLOSED BY THE INTENDED RECIPIENT TO A THIRD PARTY, REPRODUCED, COPIED, LENT, OR DISPOSED OF, DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS EXPRESSLY FURNISHED. THIS PRINT IS LENT SUBJECT TO RETURN ON DEMAND.

DRAWING NUMBER		REVISION	STATUS
12126-60-01-CS-141240		0	AFD
DATE	11/05/2020	DATE	12/05/2020	DATE	22/05/2020
DRAWN	SGH	CHECKED	ABM	APPROVED	DS

LARSEN & TOUBRO DRAWING NUMBER: 019045_080_C_V_24225_013 — REVISION: 2
OUTOTEC DRAWING NUMBER: 900707 __GAU01 __AG22__MTL01 __39825 — DOCUMENT ID: OU100680050 — REVISION: 2

LARSEN & TOUBRO
This drawing is the property of *LARSEN & TOUBRO* and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Outotec
The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية
This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

DRAWN	--	--
CHECKED	STY/ SYC/ SHH	28/05/2020
ENGINEER	--	--
APPROVED	BDA/ PRI/ JPD	30/05/2020
EMPLOYER		
SCALE	1:30	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
OXYGEN PLANT
FOUNDATION LOAD & ANCHORAGE - PIPE RACK-1

DRAWING NUMBER: MD - 216 - 3980 - EG - CV - GAD - V551-1486 — REV: D01